Check Point 3

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1 Introduction

*1. Background: Vision health is a critical component of population health, influencing quality of life, independence, and overall well-being across the lifespan. Visual impairment is associated with increased risk of injury, reduced workforce participation, and poorer mental and physical health outcomes. Disparities in vision outcomes have been observed across demographic groups, making eye health an important public health concern. Understanding how prevalence estimates vary across population characteristics can help inform prevention, screening, and access-to-care strategies.

*2. Project question: Are prevalence estimates of eye health conditions associated with age and race/ethnicity among U.S. adults?

*3. Data set overview: This analysis uses data from the National Health and Nutrition Examination Survey (NHANES) Vision and Eye Health Surveillance dataset, collected by the Centers for Disease Control and Prevention. NHANES is a nationally representative survey of the U.S. civilian, non-institutionalized population. The dataset includes demographic and vision-related prevalence estimates collected across multiple survey cycles in the United States.

2 Loading Data

```r
# Packages
library(tidyverse)

# Set working directory
setwd("/Users/lynnliu/Desktop/R Data/check point 2/")

# Load file
```

```r
nhanes_eye <- read_csv(
  "National_Health_and_Nutrition_Examination_Survey_(NHANES)_-_Vision_and_Eye_Health_Surveillance_20260
)

nhanes_eye <- nhanes_eye |>
  janitor::clean_names()

nhanes_eye_clean <- nhanes_eye |>
  filter(!is.na(data_value))

# Preview the data
glimpse(nhanes_eye)
```

Research Question and Background Research question: Are prevalence estimates of eye health conditions associated with age and demographics characteristics among U.S. adults?

*More specifically, do mean prevalence estimates of eye health conditions differ across age groups and race/ethnicity categories in the NHANES Vision and Eye Health Surveillance dataset?

*Scientific Plausibility: Eye health conditions such as age-related macular degeneration, cataracts, diabetic retinopathy, and glaucoma are strongly associated with aging. The prevalence of many ocular diseases increases substantially in older populations due to cumulative biological degeneration, chronic disease burden, and physiological changes in ocular tissues.[1]

Race and ethnicity are also associated with disparities in health outcomes in the United States. Structural inequities, differences in access to care, and social determinants of health contribute to variation in disease prevalence across racial and ethnic groups.[2]

National surveillance data demonstrate persistent disparities in vision health across demographic groups.[3] Therefore, it is scientifically plausible that prevalence estimates of eye health conditions will vary by both age and race/ethnicity among U.S. adults.

References: 1. Zhang X, Cotch MF, Ryskulova A, Primo SA, Nair P, Chou CF, Geiss LS, Barker LE, Elliott AF, Crews JE, Saaddine JB. Vision health disparities in the United States by race/ethnicity, education, and economic status: findings from two nationally representative surveys. Am J Ophthalmol. 2012 Dec;154(6 Suppl):S53-62.e1. doi: 10.1016/j.ajo.2011.08.045. PMID: 23158224; PMCID: PMC4169111. *2. Flaxman SR, Bourne RRA, Resnikoff S, Ackland P, Braithwaite T, Cicinelli MV, Das A, Jonas JB, Keeffe J, Kempen JH, Leasher J, Limburg H, Naidoo K, Pesudovs K, Silvester A, Stevens GA, Tahhan N, Wong TY, Taylor HR; Vision Loss Expert Group of the Global Burden of Disease Study. Global causes of blindness and distance vision impairment 1990-2020: a systematic review and meta-analysis. Lancet Glob Health. 2017 Dec;5(12):e1221-e1234. doi: 10.1016/S2214-109X(17)30393-5. Epub 2017 Oct 11. PMID: 29032195.* 3. Centers for Disease Control and Prevention. Vision and Eye Health Surveillance System (VEHSS). Updated 2023. Accessed February 20, 2026. https://www.cdc.gov/vision-health-data/?CDC_AAref_Val=https://www.cdc.gov/visionhealth/vehss/

#Data Preparation

Clean variable table Select variables

```r
library(dplyr)
library(tibble)
library(knitr)

# pick the 3 variables aligned with the question
vars <- c(
  "data_value",
  "age",
```

```r
  "race_ethnicity"
)

variable_overview <- tibble(variable = vars) |>
  mutate(
    class = sapply(nhanes_eye[vars], function(x) class(x)[1]),
    type = case_when(
      variable == "data_value" ~ "Continuous",
      TRUE ~ "Categorical"
    ),
    role = case_when(
      variable == "data_value" ~ "Outcome",
      TRUE ~ "Explanatory (Categorical)"
    ),
    description = case_when(
      variable == "data_value" ~ "Estimated prevalence (%)",
      variable == "age" ~ "Age group category",
      variable == "race_ethnicity" ~ "Race/ethnicity category"
    )
  )

knitr::kable(variable_overview,
             caption = "Variables included in the analysis.")
```

Table 1: Variables included in the analysis with type and analytic role.

variable	class	type	role	description
data_value	numeric	Continuous	Outcome	Estimated prevalence (%)
age	character	Categorical	Explanatory (Categorical)	Age group category
race_ethnicity	character	Categorical	Explanatory (Categorical)	Race/ethnicity category

*The outcome variable (data_value) is continuous and represents prevalence estimates (%). Age group and race/ethnicity were treated as categorical explanatory variables for grouped summary statistics and visualization. No additional discretization or transformation was required because the dataset already contains meaningful categorical groupings.

#Grouped Summary Statistics

```r
# Stratified by Age
summary_age <- nhanes_eye_clean |>
  group_by(age) |>
  summarise(
    n = n(),
    mean_prevalence = mean(data_value, na.rm = TRUE),
    sd_prevalence = sd(data_value, na.rm = TRUE),
    median_prevalence = median(data_value, na.rm = TRUE),
    IQR_prevalence = IQR(data_value, na.rm = TRUE)
  )

knitr::kable(summary_age,
             caption = "Descriptive statistics of prevalence estimates stratified by age group.")
```

```r
#Stratified by Race/Ethnicity
summary_race <- nhanes_eye_clean |>
  group_by(race_ethnicity) |>
  summarise(
    n = n(),
    mean_prevalence = mean(data_value, na.rm = TRUE),
    sd_prevalence = sd(data_value, na.rm = TRUE),
    median_prevalence = median(data_value, na.rm = TRUE),
    IQR_prevalence = IQR(data_value, na.rm = TRUE)
  )

knitr::kable(summary_race,
             caption = "Descriptive statistics of prevalence estimates stratified by race/ethnicity.")
```

#Interpretation for Age: The descriptive statistics indicate that prevalence estimates of eye health conditions vary across age groups. Mean prevalence is lowest in younger age categories (e.g., 18–39 years: 26.0%) and increases in older groups, with the highest mean observed among adults aged 80 years and older (29.8%). Median prevalence also rises across older age groups, particularly among those aged 65–79 and 80+ years.

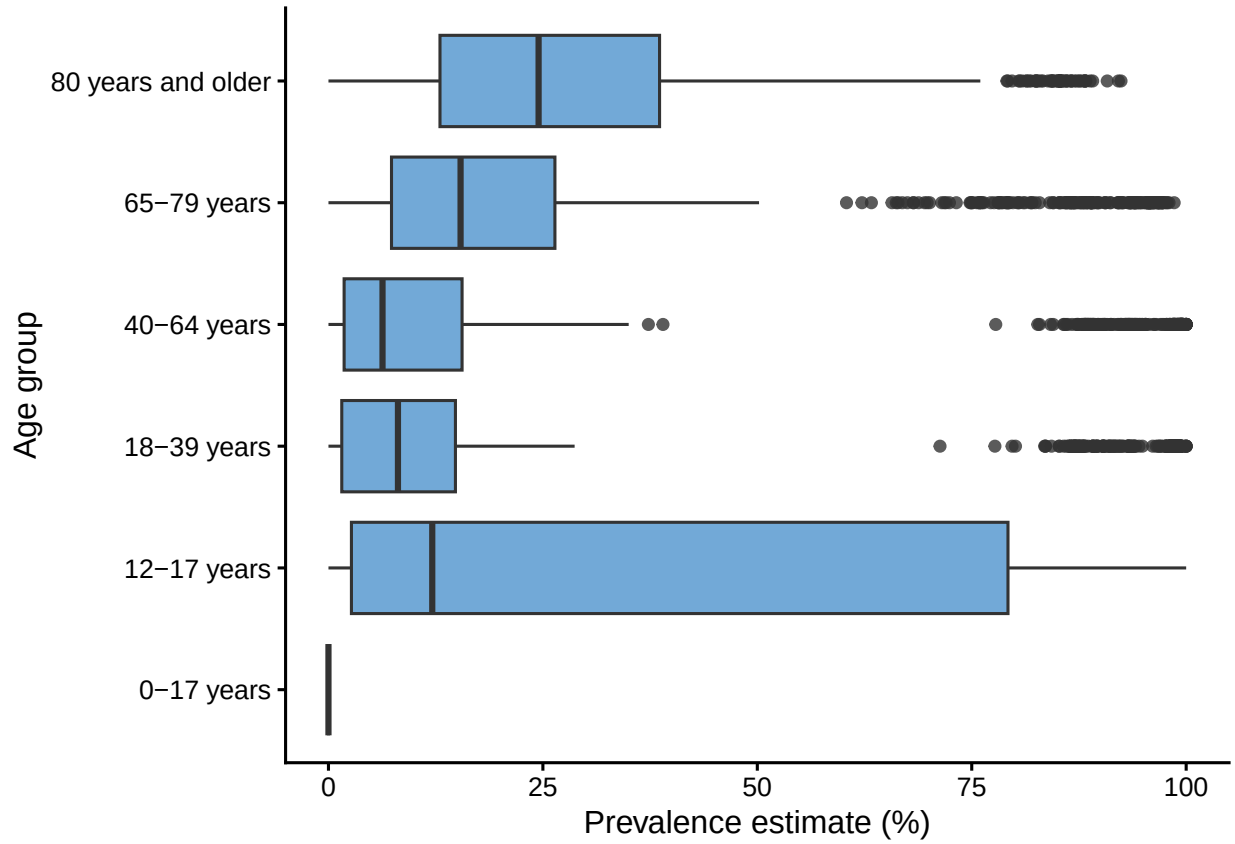
Variability, as indicated by the standard deviation and interquartile range, is substantial across several age groups, suggesting heterogeneity in prevalence estimates within age categories. Overall, these descriptive findings suggest a potential positive association between age and the prevalence of eye health conditions.

#Interpretation for Race/Ethnicity: Prevalence estimates also vary across race/ethnicity categories. The highest mean prevalence is observed in the "Other" race category (28.5%), followed by Black non-Hispanic (25.1%) and Hispanic (24.6%) populations. White non-Hispanic individuals have a lower mean prevalence (22.2%) relative to several other groups.

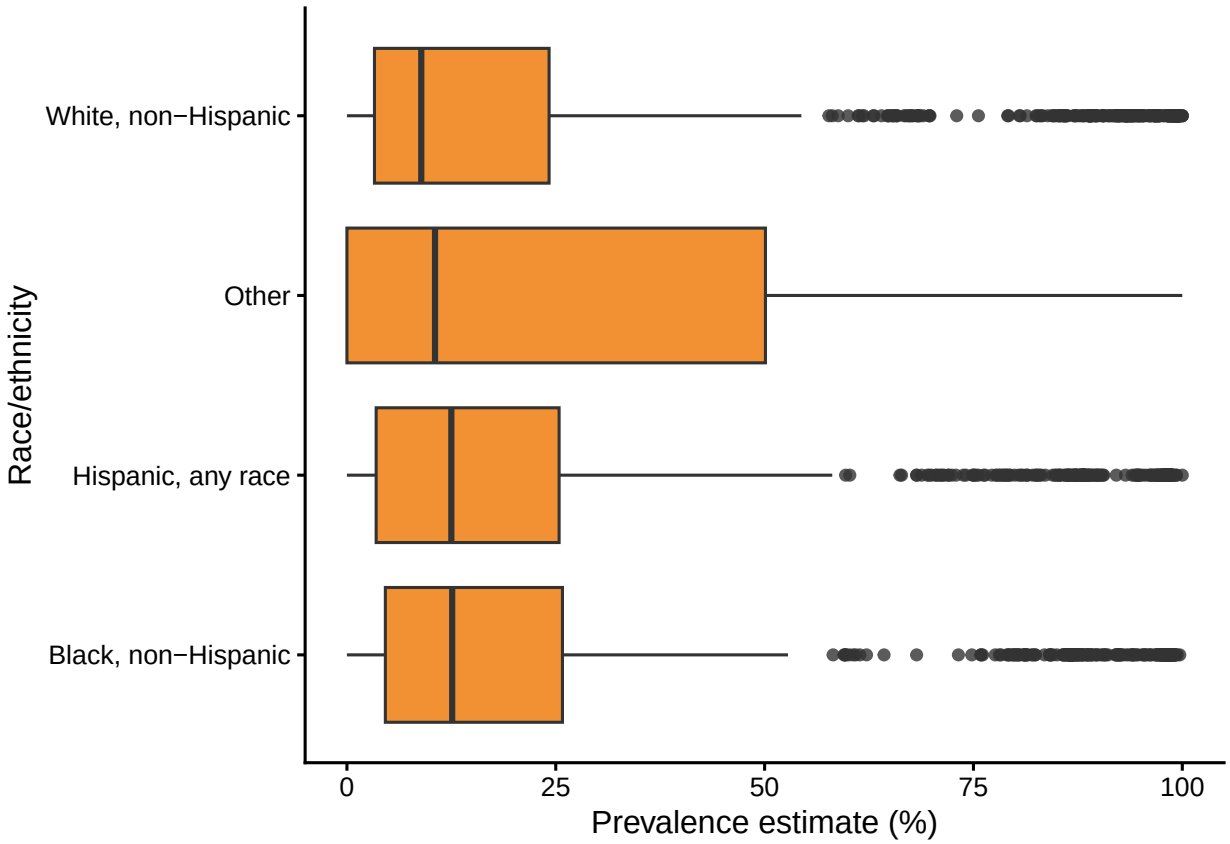
The standard deviation and IQR values indicate considerable variability within racial/ethnic groups, particularly in the "Other" category, which shows the largest spread (IQR = 50.1). These differences suggest potential disparities in eye health prevalence across socioeconomic and demographic groups.

3 Compartive Visualizations

```r
#Prevalence by Age
library(ggplot2)

p_age <- ggplot(
  nhanes_eye_clean |> filter(age != "All ages"),
  aes(x = age, y = data_value)
) +
  geom_boxplot(fill = "steelblue3", alpha = 0.8) +
  coord_flip() +
  labs(
    x = "Age group",
    y = "Prevalence estimate (%)"
  ) +
  theme_classic(base_size = 12)

p_age
```




```
ggsave("figure_prevalence_by_age.pdf",
       plot = p_age,
       width = 6,
       height = 4,
       device = "pdf")

#Prevalence by Race/Ethnicity
p_race <- ggplot(
  nhanes_eye_clean |> filter(race_ethnicity != "All races"),
  aes(x = race_ethnicity, y = data_value)
) +
  geom_boxplot(fill = "darkorange2", alpha = 0.8) +
  coord_flip() +
  labs(
    x = "Race/ethnicity",
    y = "Prevalence estimate (%)"
  ) +
  theme_classic(base_size = 12)

p_race
```



#Interpretation for age: The boxplot shows increasing central tendency and distribution of prevalence estimates across older age groups. Median values are higher among adults aged 65 years and older, supporting a potential positive association between age and eye health prevalence

#Interpretation for Race/Ethnicity: Differences in median and distribution of prevalence estimates are observed across race/ethnicity groups. Some groups display higher central tendencies and greater variability, suggesting potential disparities in vision health prevalence. These patterns are consistent with descriptive summary statistics.

4 AI Use Disclosure Statement

As part of this assignment, please indicate whether you used any AI-based tools (e.g., ChatGPT, Claude, Copilot, Gemini, etc.). Indicate:

- *Did you use AI? Yes / No* YES
- *If yes: Write a short disclosure statement (2–3 sentences) describing:*
- *Which tool(s) you used.*
- *How you used it (e.g., to help decide on an analysis approach, to generate a first draft of code, to improve or debug code you had written yourself, etc.).*

Example: "This document was generated using Claude to generate original code, which was then reviewed and adjusted" or "This document was generated using ChatGPT to assist with debugging of code generated by the author".

Please keep your statement brief and honest. The goal is transparency, not detail: we do not need exact prompts or transcripts, just a clear sense of how AI supported this work.

This document was generated using ChatGPT to assist with understanding the assignment instructions and drafting initial R code for data subsetting, visualization, and probability calculations.